CS2 Week 5 Annotated

DOFB with Integral Action, Intro to MIMO systems

Exercise Classes

- Shared exercise with Haixiao and Kissan
- Theory recap
- Notes available on Kissan's website: https://kissanv.github.io

& on Haixiao's Polybox: https://polybox.ethz.ch/index.php/s/AReA9YBeJQKxBDn

- Quizzes, Examples & Old exam questions

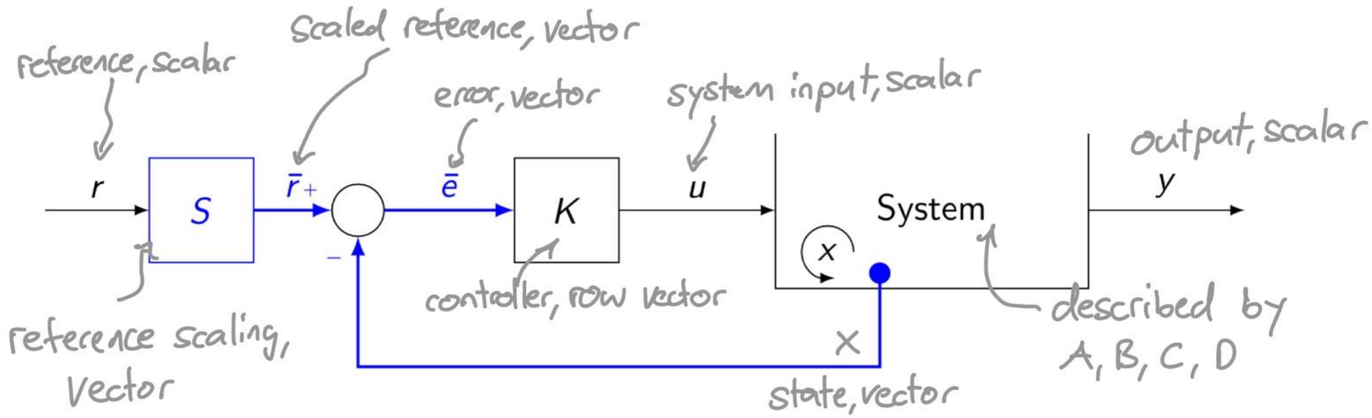
Kissan's Website

Haixiao's Polybox

Recap

State Feedback

- assume that we know the states x → smarter to directly feed back x instead of y
- the control architecture would look like this:



LQR

method to find optimal controller K for a given task

$$\min J(x,u) = \int_0^\infty [x(t)^\top Q x(t) + u(t)^\top R u(t)]dt$$

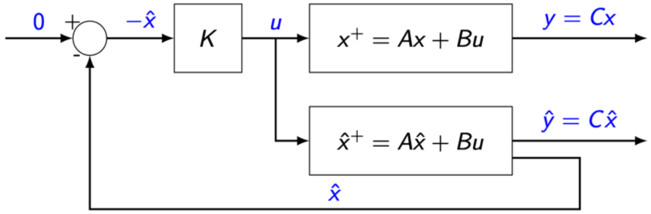
$$\text{s.t.} \quad \dot{x}(t) = Ax(t) + Bu(t)$$
$$u(t) = -Kx(t)$$

Solution: $A^\top P + PA + Q - PBR^{-1}B^\top P = 0$

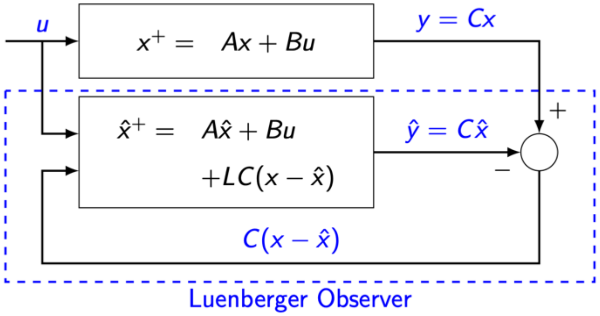
→ Solve this equation for P (positive definite and symmetric!)
→ The optimal controller is calculated by: $K = R^{-1}B^\top P$

LQE

- We estimate the state x instead of measuring it directly with sensors

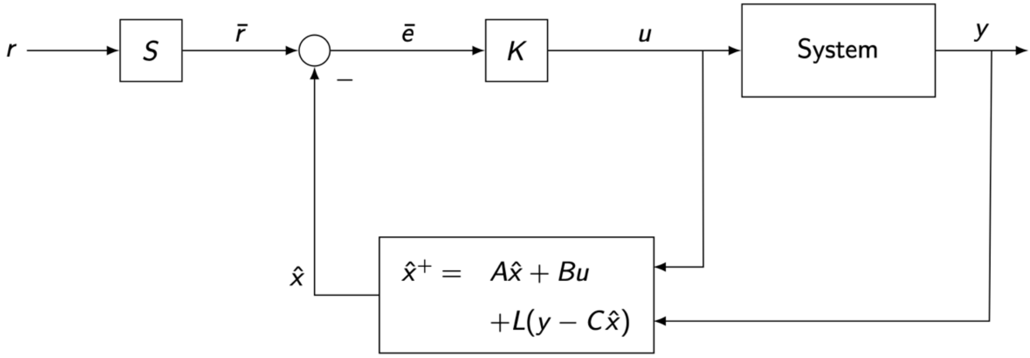


- Compare the measured output $y = Cx$ with the predicted output $C\hat{x}$
- Use their difference to correct the state estimate $\hat{x}$



- Want to minimize: $\lim_{t \to \infty} E[(x - \hat{x})(x - \hat{x})^T]$
- Solution: $YC^T R^{-1} AY + YA^T - YC^T R^{-1} CY + Q = 0$

DOFB



- Idea: state feedback with estimated states

- The closed loop system has the following form:

$$\boxed{\begin{aligned} x^+ &= Ax - BK\hat{x} + BKSr \\ \widehat{x^+} &= (A - LC - BK)\hat{x} + BKSr + LCx \\ y &= Cx \end{aligned}}$$

$$\xrightarrow[\text{\& rewriting}]{\text{introducing: } \eta = x - \hat{x}} \quad \begin{bmatrix} x \\ \eta \end{bmatrix}^+ = \begin{bmatrix} A - BK & BK \\ 0 & A - LC \end{bmatrix} \begin{bmatrix} x \\ \eta \end{bmatrix}$$

$\Rightarrow$ the poles are given by $\det(sI - A_{cl}) = \det\big(sI - (A - BK)\big)\det\big(sI - (A - LC)\big)$

Based on seperation principle we can

- Find K with LQR
- Find L with LQE

$\Big\}$ combine them to LQG $\rightarrow$ system has optimal ctrl. / obs. gain!

Quiz

Which of the following statements about state estimation in state feedback control for LTI systems are true?

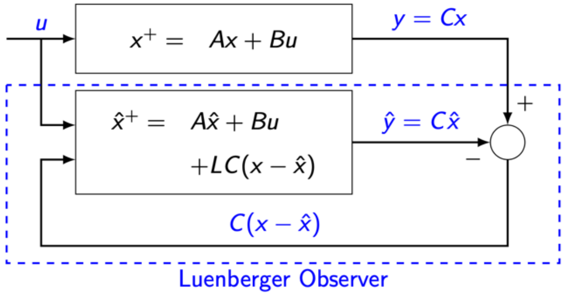
A. To design a Luenberger observer for a state feedback controller, additional sensors must be installed to measure all the states in the system.

B. A Luenberger observer is sensitive to measurement noise, where high observer gains magnify measurement noise.

C. The controller and the estimator must be designed together because they interact with each other.

D. If the observer gain is chosen large enough, measurement noise will automatically disappear.

Quiz



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Course Schedule

Subject	Week
Intro – Discrete Time Systems	1
Diagonalization, Modal Coordinates, Contr/Obs.	2
State Feedback, Pole Placement, LQR	3
State Estimation, Luenberger Obs., Seperation Principle, LQG	4
DOFC with integral action, Intro to MIMO systems	5
Nonlinear Systems	6
Nonlinear Systems	7
break	8
Discrete-time Unconstrained Optimal Control	9
Convex Optimization	10
Model Predictive Control	11
Practical Model Predictive Control	12
Robust Model Predictive Control	13
Implementation Methods	14

Today

1. DOFB with integral action
2. Intro MIMO systems
3. Norms

1. DOFC w/ integral action

DOFC with integral action

- to eliminate steady-state error, we introduce integral state (same as LQR servo)

$$x_I = \int (r-y)dt \Rightarrow \dot{x}_I = r - y = r - Cx$$

- Now we augment the plant to include this state:

Previously

$\dot{x} = Ax + Bu$

$y = Cx$

Augmenting

$x \rightarrow \begin{bmatrix} x \\ x_I \end{bmatrix} = x'$

$$\begin{bmatrix} \dot{x} \\ \dot{x}_I \end{bmatrix} = \overbrace{\begin{bmatrix} A & 0 \\ -C & 0 \end{bmatrix}}^{A'} \begin{bmatrix} x \\ x_I \end{bmatrix} + \underbrace{\begin{bmatrix} B \\ 0 \end{bmatrix}}_{B'} u + \begin{bmatrix} 0 \\ I \end{bmatrix} r$$

$$y = \underbrace{\begin{bmatrix} C & 0 \end{bmatrix}}_{C'} \begin{bmatrix} x \\ x_I \end{bmatrix}$$

For A', B', C' we can find a new stabilizing feedback gain

$K' = \begin{bmatrix} K & K_I \end{bmatrix}$, s.t. $u = -K'x' + KSr \Rightarrow u = -Kx - K_I x_I + KSr$

- x: unknown! $\Rightarrow$ Use estimate $\hat{x}$
- x_I: known! $\Rightarrow$ Only requires y and r

LQG stability margins

Using this feedback law and again the coordinate change $\eta = x - \hat{x}$:

$$\begin{bmatrix} x \\ x_I \\ \eta \end{bmatrix}^+ = \begin{bmatrix} A - BK & -BK_I & BK \\ -C & 0 & 0 \\ 0 & 0 & A - LC \end{bmatrix} \begin{bmatrix} x \\ x_I \\ \eta \end{bmatrix} + \begin{bmatrix} BKS \\ I \\ 0 \end{bmatrix} r \qquad y = \begin{bmatrix} C & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ x_I \\ \eta \end{bmatrix}$$

Recall: If the conditions for state-feedback are given, then the LQR procedure always finds a stabilizing and robust controller
Also: Robustness can be measured with phase/gain margins

$\Rightarrow$ LQR guarantees:

- phase margin $\geq 60°$
- gain margin ≥ 6 dB

Do we still guarantee robustness with an optimal observer (LQG)?

NO guaranteed stability margins for LQG

Summarized

Reference scaling with feedforward

- $u = K(Sr - \hat{x}) = \bar{N}r - K\hat{x}$
- Fast nominal tracking
- Works well if the model is accurate
- No integral correction of constant bias

Feedback with integral action

- $u = -K\hat{x} - K_I x_I$
- $x_I^+ = r - y$
- Integral action removes steady-state step errors
- More robust to disturbances and model mismatch
- Can be sluggish

Feedforward + feedback

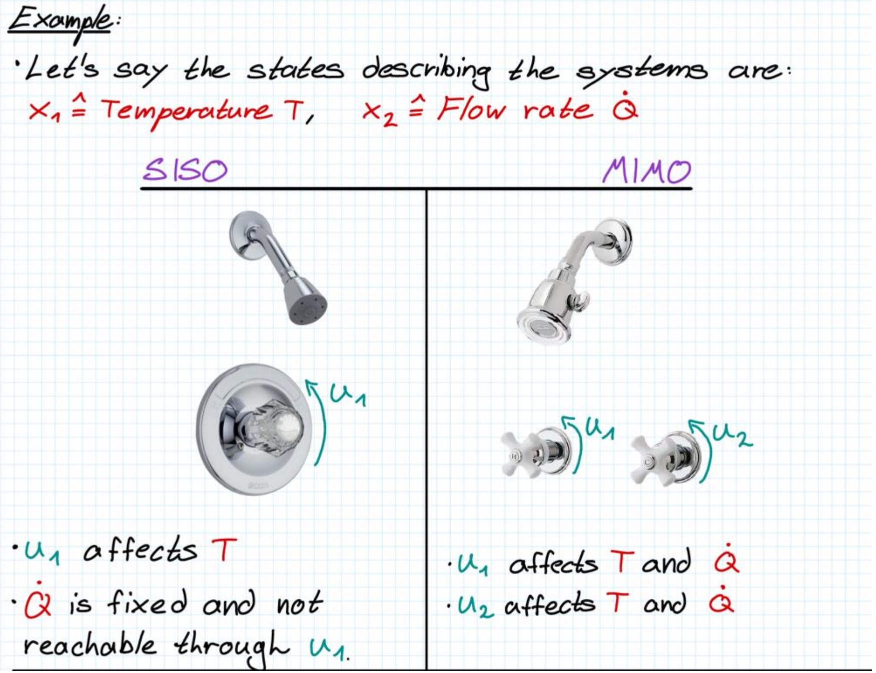
- $u = \bar{N}r - K\hat{x} - K_I x_I$
- Prompt nominal action from $\bar{N}r$
- Feedback corrects disturbances and model errors
- Often the best practical compromise

2. Intro MIMO systems

Introduction

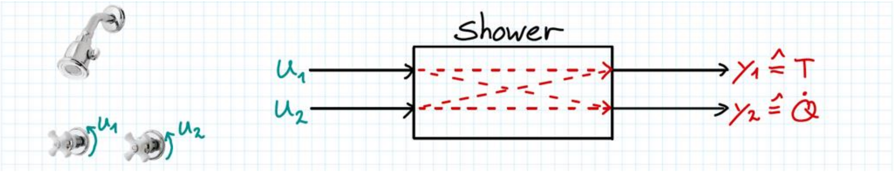
- Up to now, we have assumed all analyzed systems to be controlled by only one input and measured by only one output → SISO
- In reality, many systems have multiple inputs and multiple outputs
- To get a feeling for how MIMO systems affect complexity, let's gain some intuition

- In a MIMO system, we need specific combinations of inputs to obtain a desired output



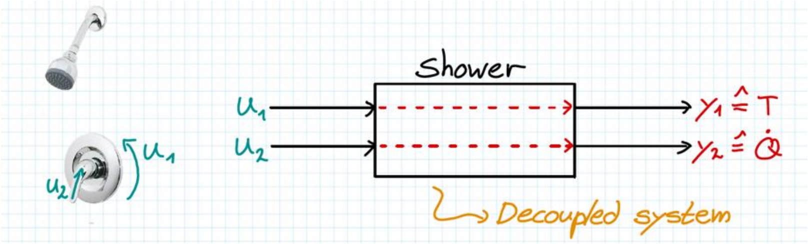
Introduction

- The previous example visualizes the interconnectivity of MIMO systems



- This is the general case for MIMO systems

- In systems where each input affects only one separate output, we call the MIMO system decoupled



- The transfer function representation makes the system's internal connections visible, since it relates each input to each output.

Introduction

But how can we write a transfer function for a MIMO system?

- Recall how we wrote transfer functions for SISO systems:

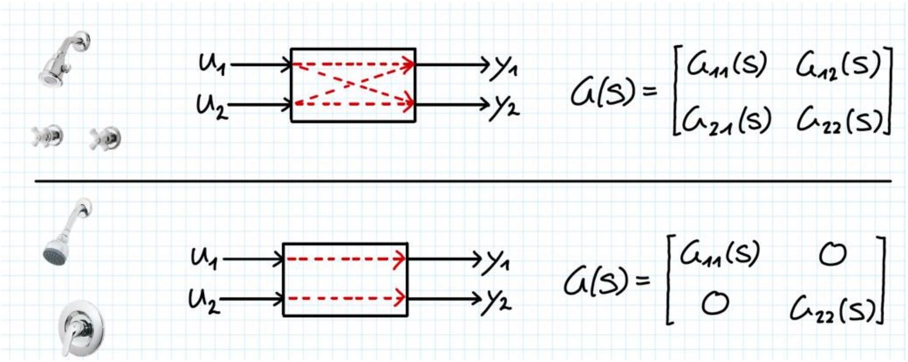
$$G(s) = \frac{Y(s)}{U(s)} = \frac{\prod_i (s - z_i)}{\prod_i (s - p_i)}$$

→ Output to input ratio

- For MIMO systems, each output has a ratio to each input
- Instead of scalar transfer functions, we now have transfer function matrices.

$$G(s) = \begin{bmatrix} G_{11}(s) & \cdots & G_{1m}(s) \\ \vdots & \ddots & \vdots \\ G_{l1}(s) & \cdots & G_{lm}(s) \end{bmatrix}$$

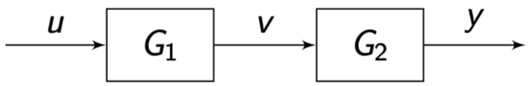
l: number of outputs
m: number of inputs
$\Rightarrow G(s) \in \mathbb{C}^{l \times m}$



MIMO block diagrams

- There is a crucial difference when dealing with MIMO block diagrams

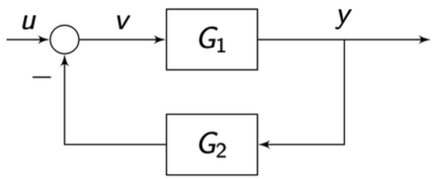
⇒ In general, matrices do not commute!



$\cancel{y = G_1 G_2 u}$ wrong

$y = G_2 G_1 u$ ✓

- The following feedback loop highlights additional rules:



$y = G_1 v, \quad v = u - G_2 y$

$y = G_1(u - G_2 y) = G_1 u - G_1 G_2 y$

$\rightsquigarrow y = (I + G_1 G_2)^{-1} G_1 u$

General rules (same as CS1):

- Start from output and proceed against signal flow.
- Multiply inverse on the side where the term that you want gone is.

For the derivation of more complex systems, the following rule can be handy: "Push-through identity"

$$\boxed{G_1(I + G_2 G_1)^{-1} = (I + G_1 G_2)^{-1} G_1}$$

MIMO state-space

- The system dynamics can be written as:

$$x^+ = \underbrace{\overbrace{A}^{n\times n}\overbrace{x}^{n\times 1}}_{n\times 1} + \underbrace{\overbrace{B}^{n\times \mathbf{m}}\overbrace{u}^{\mathbf{m}\times 1}}_{n\times 1},$$

$$y = \underbrace{\overbrace{C}^{\ell\times n}\overbrace{x}^{n\times 1}}_{\ell\times 1} + \underbrace{\overbrace{D}^{\ell\times \mathbf{m}}\overbrace{u}^{\mathbf{m}\times 1}}_{\ell\times 1}.$$

- $x \in \mathbb{R}^n$, where n is the **order** of the system.
- $u \in \mathbb{R}^m$, where m is the number of inputs.
- $y \in \mathbb{R}^\ell$, where ℓ is the number of outputs.

- Nothing changes except for dimensionalities!

- state-space ⟶ transfer function : $\boxed{G(s) = C(sI - A)^{-1}B + D}$

- So much for the basics. To analyze how a system behaves, we must look at its poles and zeros. But how do we do that if the transfer function is a matrix?

Poles & Zeros

- The general meaning of poles and zeros remains the same

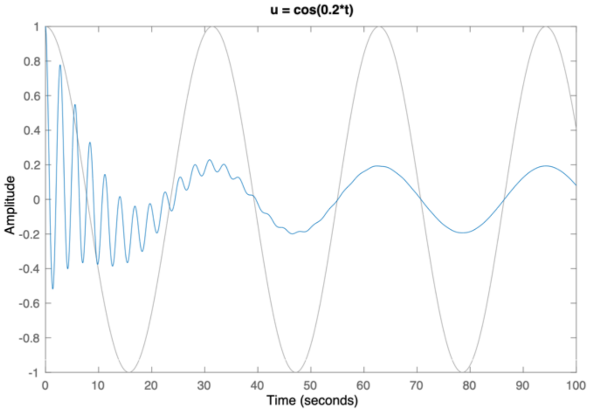
⇒ pole: "Generated frequency" at the output, regardless of the input

⇒ zero: "Absorbed frequency" that is not transmitted from the input to the output

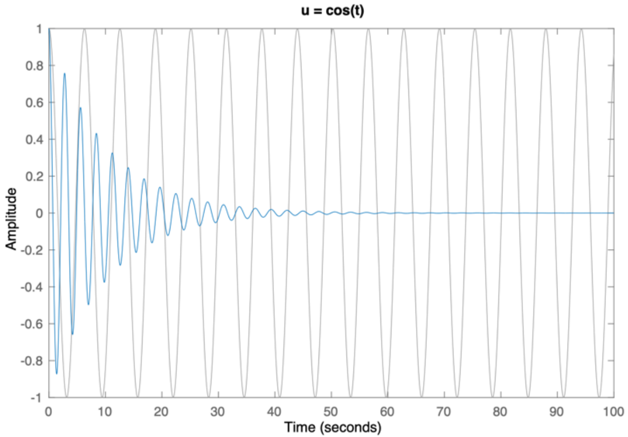
Let's look at an example from a SISO system: $G(s) = \dfrac{s^2 + 1}{s^2 + 0.2s + 5}$

pole: $\pi = -0.1 \pm 2.23\, i$

zero: $\zeta = 1$



Pole frequency appears in the output independent of the input



Pole effect remains in the transient, while the zero cancels the input frequency

MIMO poles

Definition: MIMO poles are the roots of the pole polynomial of the least common multiple of the denominators of all possible minors of the transfer function matrix.

What are minors again?

Example: let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

1x1 minors: a, b, c, d

2x2 minors: $\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - cb$

minors are the determinants of a square sub-matrix

With this we can find the poles of a MIMO system…

Example

Let $G(s) = \begin{bmatrix} \frac{s+2}{(s+3)^2} & 0 \\ 0 & \frac{s}{(s+2)(s+3)} \end{bmatrix}$

Now we write down all minors:

$1\times1: \frac{s+2}{(s+3)^2}, \frac{s}{(s+2)(s+3)}$ $\quad 2\times2: \det G(s) = \frac{\cancel{s+2}}{(s+3)^2} \cdot \frac{s}{\cancel{(s+2)}(s+3)} = \frac{s}{(s+3)^3}$

Find LCM of all minor denominators:

$d(s) = (s+2)(s+3)^2$

The roots of $d(s)$ are the poles:

$\pi_1 = -2 \quad \pi_2 = -3$ (multiplicity $m=3$)

Simple example since $G(s)$ is decoupled; for complex systems, cancellations and multiplicities are less obvious.

Note: *Stability*
With poles we can analyze the stability of MIMO systems
⇒ Same conditions as for SISO:

- Stability: $Re(p_i) \leq 0$ (in CT)
- Asymptotic stability: $Re(p_i) < 0$ (in CT)

MIMO zeros

- MIMO zeros are a little more complicated…
- We differentiate between two types:

  - Transmission zeros
  - Invariant zeros

To properly understand their difference we look at them from different perspectives

Transmission zero definition <u>(s-domain perspective)</u>:

$s = s_0$ is a transmission zero of system $G(s)$ if
∃ a non-zero rational function $u_0(s)$, s.t. $lim_{s \to s_0} G(s)\, u_0(s) = 0$
equivalently: $G(s_0)$ has a non−trivial nullspace (rank loss)

Hard to gain an intuitive understanding, but in essence we now have combinations of inputs (directions) that make the output go to zero.

Example: decoupled

$lim_{s \to s_0} G(s)\, u_0(s) = 0$

· Assume $G(s) = \begin{bmatrix} \frac{s+2}{(s+3)^2} & 0 \\ 0 & \frac{s}{(s+2)(s+3)} \end{bmatrix}$

The input vector will have structure $u = [u_1, u_2]^T$

If we now specifically choose to only excite u_1, i.e. $u_0 = [1, 0]^T$:

$$G(s)\,u_0(s) = \begin{bmatrix} \frac{s+2}{(s+3)^2} & 0 \\ 0 & \frac{s}{(s+2)(s+3)} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{s+2}{(s+3)^2} \\ 0 \end{bmatrix}$$

for $s = -2$, this approaches 0

⇒ For this specific combination (i.e., direction) of inputs, we have a zero at $s = -2$.

Note: There is also a pole at $s = -2$, but they do not cancel!

Example: coupled & unintuitive

$\lim_{s \to s_0} G(s)\, u_0(s) = 0$

- Assume $G(s) = \begin{bmatrix} 1 & \frac{1}{s+1} \\ 0 & 1 \end{bmatrix}$

There's a pole at -1!

But are there zeros?

Let's apply the definition:

$$\lim_{s \to s_0} \begin{bmatrix} 1 & \frac{1}{s+1} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \lim_{s \to s_0} \begin{bmatrix} u_1 + \frac{u_2}{s+1} \\ u_2 \end{bmatrix} = 0$$

$\Rightarrow$ This eqn. holds for $u = [-1, (s+1)]^T \rightsquigarrow$ zero at -1

$\Rightarrow$ Thus $\lim_{s \to -1} [G(s) u_0(s)] = \lim_{s \to -1} \begin{bmatrix} -1+1 \\ s+1 \end{bmatrix} = 0$

Conclusion: There is a pole and a zero at $s = -1$, but they do not cancel because they exist in different directions.

Invariant zeros

To understand invariant zeros, we look at our systems from the state-space perspective:

- Assume we have a (stable) system: $\dot{x} = Ax + Bu$, $y = Cx + Du$

1. Steady-state is described by: $x(t) = (sI - A)^{-1} B u_0 e^{st}$

   with $x(t) = x_0 e^{st}$ $\rightarrow$ $(sI - A)x_0 - Bu_0 = 0$ (1)

2. The output is given by: $y(t) = Cx_0 e^{st} + Du_0 e^{st}$

   $\Rightarrow$ We want to find zeros! Therefore we set $y(t) = 0$ $\rightsquigarrow$ $Cx_0 \cancel{e^{st}} + Du_0 \cancel{e^{st}} = 0 \rightarrow Cx_0 + Du_0 = 0$ (2)

Combining (1)+(2): $\begin{bmatrix} sI - A & -B \\ C & D \end{bmatrix} \begin{bmatrix} x_0 \\ u_0 \end{bmatrix} = 0$

Important: If the realization (A, B, C, D) is minimal, then invariant zeros ≙ transmission zeros

- If this has a non-trivial solution (x, u), then the output y is zero even though the input (and state) are not zero!
- The values of s for which this matrix becomes singular are the invariant zeros of the system!

Reachable Cannonical Form

- What we have not looked at is how to find a state-space realization given a tf matrix
- What we know already: given the MIMO state-space system (A, B, C, D), we find the tf matrix as: $G(s) = C(sI - A)^{-1}B + D$

But what about the other way? i.e., tf → SS

SISO:

$$C(sI - A)^{-1}B + D = \frac{c_{n-1}s^{n-1} + c_{n-2}s^{n-2} + \ldots + c_0}{s^n + a_{n-1}s^{n-1} + \ldots + a_0} + d. \longrightarrow$$

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & \ldots & 0 \\ 0 & 0 & 1 & 0 & \ldots & 0 \\ \vdots & & & \ddots & & 1 \\ -a_0 & -a_1 & \ldots & & & -a_{n-1} \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}$$

$$C = \begin{bmatrix} c_0 & c_1 & \ldots & c_{n-1} \end{bmatrix}, \quad D = [d];$$

MIMO:

$$G(s) = \begin{bmatrix} G_{11}(s) & \cdots & G_{1m}(s) \\ \vdots & \ddots & \vdots \\ G_{l1}(s) & \cdots & G_{lm}(s) \end{bmatrix}$$

calculate a canonical form for each G_{ij}?

$$A = \begin{bmatrix} A_{11} & & & & & \\ & A_{12} & & & & \\ & & \ddots & & & \\ & & & A_{21} & & \\ & & & & \ddots & \\ & & & & & A_{\ell m} \end{bmatrix}, \quad B = \begin{bmatrix} b_{11} & & \\ & b_{12} & \\ & & \ddots \\ b_{21} & & \\ & \vdots & \\ & & b_{\ell m} \end{bmatrix}$$

$$C = \begin{bmatrix} c_{11} & c_{12} & \ldots & & & & \\ & & & c_{21} & \ldots & & \\ & & & & & \ddots & \\ & & & & & c_{\ell 1} & \ldots & c_{\ell m} \end{bmatrix}, \quad D = \begin{bmatrix} d_{11} & \cdots & d_{1m} \\ \vdots & \ddots & \vdots \\ d_{\ell 1} & \cdots & d_{\ell m} \end{bmatrix}$$

Good news: This is valid!

Bad news: This realization is not minimal

Gilbert's Realization

- A much better method relies on trying to figure out the minimum number of "copies" of each pole that we need to construct a realization of a MIMO transfer function.
- The idea is the following. Write the transfer function as

$$G(s) = \frac{H(s)}{d(s)} + D,$$

where $d(s)$ is the least common denominator of all the entries in $G(s)$, and $D = \lim_{s\to\infty} G(s)$ is the feed-through term.
- If $d(s)$ has no repeated roots, one can use Gilbert's method to construct a realization of $G(s)$. The first step is to write a (matrix) partial fraction expansion of $H(s)/d(s)$, i.e., write

$$G(s) = \frac{H(s)}{d(s)} + D = \frac{R_1}{s-p_1} + \frac{R_2}{s-p_2} + \ldots + \frac{R_{n_d}}{s-p_{n_d}} + D,$$

where n_d is the degree of $d(s)$, p_i, $i = 1 \ldots n_d$ are the roots of $d(s)$, and

$$R_i = \lim_{s\to p_i} (s-p_i)G(s).$$

- Let r_i be the rank of the matrix R_i, for each $i = 1, \ldots, n_d$; this indicates the number of poles at location p_i that are needed in the realization.
- The order of the resulting state-space model would be $n = \sum_{i=1}^{n_d} r_i \geq n_d$.
- Write $R_i = C_i B_i$ for some appropriate matrices B_i, C_i, such that
  - B_i has as many columns as inputs to the system,
  - C_i has as many rows as outputs of the system, and
  - B_i and C_i have r_i independent rows/columns, respectively.
- Finally, assemble the state-space model as

$$A = \begin{bmatrix} p_1 I_{r_1\times r_1} & & & \\ & p_2 I_{r_2\times r_2} & & \\ & & \ddots & \\ & & & p_{n_d} I_{r_{n_d}\times r_{n_d}} \end{bmatrix}, \quad B = \begin{bmatrix} B_1 \\ B_2 \\ \vdots \\ B_{n_d} \end{bmatrix}$$

$$C = \begin{bmatrix} C_1 & C_2 & \cdots & C_{n_d} \end{bmatrix}, \quad D.$$

Example: Find a ss-representation of $G(s) = \begin{bmatrix} 1 & \frac{1}{s+1} \\ 0 & 1 \end{bmatrix}$

1) $D = \lim_{s\to\infty} G(s)$ $\quad D = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

2) Find $R_i = \lim_{s\to p_i} (s-p_i)G(s)$ we have $p_1 = -1$

$$R_1 = \lim_{s\to -1} (s+1)\begin{bmatrix} 1 & \frac{1}{s+1} \\ 0 & 1 \end{bmatrix} = \lim_{s\to -1} \begin{bmatrix} s+1 & 1 \\ 0 & s+1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

3) Find $R_i = C_i B_i$

$$R_1 = \underbrace{\begin{bmatrix} 1 \\ 0 \end{bmatrix}}_{C_1} \underbrace{\begin{bmatrix} 0 & 1 \end{bmatrix}}_{B_1}$$

also $r_1 = \text{rank}(R_1) = 1$

4) Find A by setting $A = \begin{bmatrix} p_1 \mathbb{I}_{r_1\times r_1} & & \\ & \ddots & \\ & & p_n \mathbb{I}_{r_n\times r_n} \end{bmatrix}$ $\rightarrow$ $p_1 = -1,\ r_1 = 1$

hence $A = [-1]$

5) Putting everything together

$$A = [-1] \quad B = [0 \ \ 1] \quad C = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad D = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Quick analysis

- From A we immediately see the pole at $s=-1$
- For the zeros we can construct $\begin{bmatrix} sI-A & -B \\ C & D \end{bmatrix}$

$$\begin{bmatrix} s+1 & 0 & -1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \rightsquigarrow \text{becomes singular for } s=-1$$

Conclusion: we have $p=-1$ & $\xi=-1$

3. Norms

Signal amplification in MIMO sytems

- For many performance criteria we had to look at the amplification of signals in the SISO case (recall Bode plots)
- We did this by analyzing the magnitudes of transfer functions

Now, how do we translate this to MIMO systems?

- Let's illustrate the added complexity, assume $y = Gu$ with:

$$G = \begin{bmatrix} 0 & 10 \\ 1 & 1 \end{bmatrix}$$

- if we choose $u = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, then $y = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$
- if we choose $u = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, then $y = \begin{bmatrix} 10 \\ 1 \end{bmatrix}$

- For the same size input we get very different output "sizes"
- We call this directionality of MIMO systems
- To quantify amplification or "size" we need new tools → norms

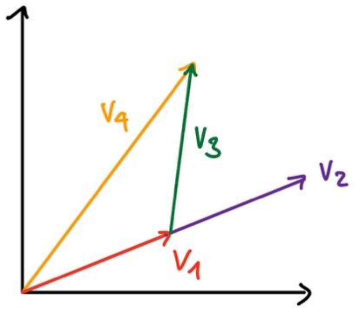
Vector norms

- Measures the "length" of a vector
- In higher dimensions, this notion becomes less intuitive

⇒ We need a precise mathematical definition of "length"

A norm maps vectors to a non-negative scalar with these properties:

- Positivity: $||x|| > 0 \ \forall x \neq 0$
- Homogeneity: $||a\,x|| = ||a||\,||x||, \ \forall a \in \mathbb{R}, x \in V$
- Triangle inequality: $||x + y|| \leq ||x|| + ||y||$



1) $||v_1|| > 0$

2) $v_2 = 2 \cdot v_1$

$||v_2|| = ||2 \cdot v_1|| = 2 \cdot ||v_1||$

3) $||v_1 + v_3|| = ||v_4|| < ||v_1|| + ||v_3||$

Vector norm examples:

- Euclidean norm:

$||x|| = \sqrt{x'x}$; $x' \triangleq$ conjugate transpose

- p-norm:

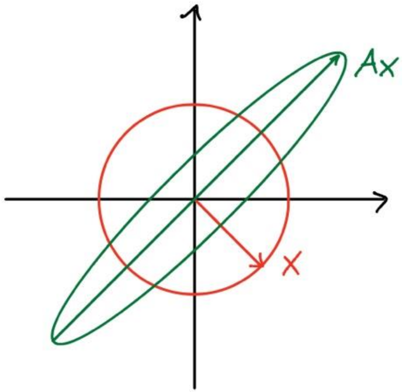
$||x||_p = \left(\sum_{i=1}^{n} x_i^p \right)^{1/p}$ ($p=2 \rightsquigarrow$ euclidean norm)

Matrix norms

- Most relevant matrix norms belong to a subclass called induced norms
- They measure how "large" a matrix is based on how much it amplifies a vector

- Induced p-norm:

$$\|A\|_p := \sup_{x \neq 0} \frac{\|Ax\|_p}{\|x\|_p} = \max_{\|x\|_p = 1} \|Ax\|_p$$



- The are some important matrix norms that are not induced, e.g.:

  - Frobenius norm:

$$\|A\|_F := \left(\sum_{i=1}^{m} \sum_{j=1}^{n} |a_{ij}|^2 \right)^{1/2} = \left[\mathrm{Trace}(A'A) \right]^{1/2}$$

Feedback?

Too fast? Too slow? Less theory, more exercises?

I would appreciate your feedback. Please let me know.